

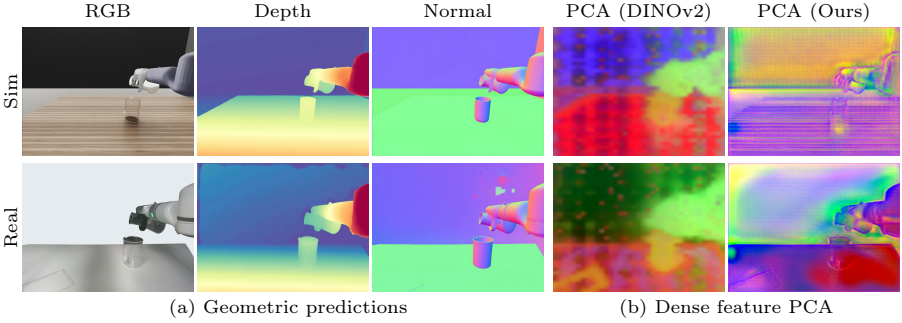


Fig. 3: Visualization of domain invariance. (a) Predicted depth and normals are visually consistent between sim and real. (b) PCA of dense features: DINOv2 shows distinct color patterns across domains; our geometric encoder features are well aligned.

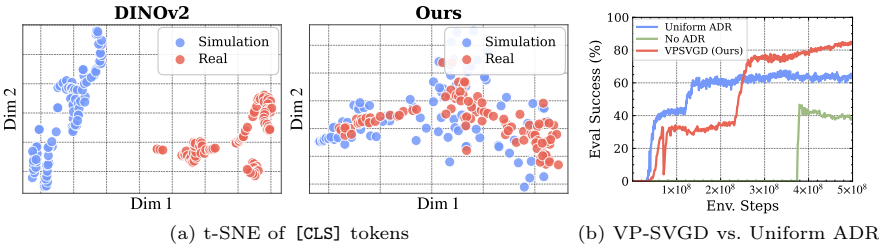


Fig. 4: Feature analysis and training efficiency. (a) t-SNE of [CLS] tokens: DINOv2 tokens cluster by domain; our encoder tokens are well mixed, confirming domain-invariant representations. (b) VP-SVGD achieves higher final success with fewer environment steps than Uniform ADR by concentrating updates on the difficulty frontier.

namics. Without ADR, the full parameter range is exposed from the start, making early training difficult and convergence slow. Uniform ADR eases the initial phase by starting from a narrow range and expanding gradually, but it allocates budget uniformly across all dimensions, including already-mastered ones; reaching a comparable final performance therefore requires significantly more training steps. VP-SVGD balances the difficulty across parameter dimensions by concentrating updates on the frontier where the policy currently struggles, achieving both faster convergence and higher final success with fewer environment steps. Full details of randomization bounds are in Appendix B.1.

Real-world qualitative results. Fig. 5 shows keyframe sequences from successful real-world trials. Two observations stand out. First, the policy reliably grasps transparent beakers that are nearly invisible to commodity depth sensors, confirming that the geometric encoder extracts sufficient shape information from RGB alone. Second, the pouring and handover sequences exhibit smooth, temporally coherent motions without the oscillations typical of single-step policies, which we attribute to the action-chunk formulation of the diffusion policy.

Failure analysis. We observe two failure modes with distinct root causes. (1) *Perception*: at extreme object poses near the training boundary, the encoder

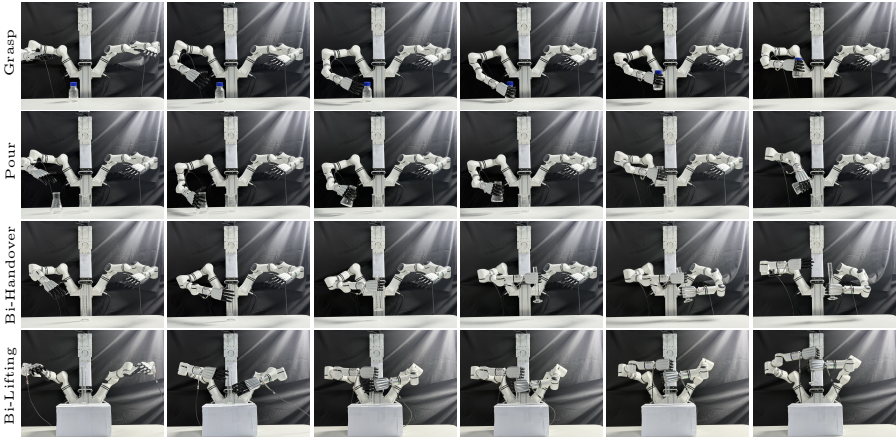


Fig. 5: Real-world experiments. Each row shows keyframes from a successful trial.

underestimates depth discontinuities at the beaker rim, causing grasp aperture misjudgment. The same object at moderate poses is grasped reliably, indicating that viewpoint diversity in the training renders, rather than the supervision signal, is the bottleneck. (2) *Control*: in Bimanual Handover, the left hand occasionally releases before the right hand establishes a stable grasp. Both hands share a single action chunk, so release timing must be learned implicitly from data; an explicit contact-conditioned gating mechanism may alleviate this.

5 Conclusion

We presented GEARS, a framework for zero-shot sim-to-real dexterous manipulation coupling geometric perception with generative policy transfer. A geometric encoder fine-tuned with joint depth and normal prediction provides domain-invariant features; VP-SVGD steers domain randomization toward difficult regions for efficient expert training; and a diffusion policy distills the expert’s multimodal behaviors into temporally coherent action chunks. GEARS averages 67.5% success on four real-world tasks involving transparent and reflective objects, outperforming the strongest baseline by 25 points. Ablations reveal a super-additive interaction between the geometric encoder and the diffusion policy, confirming that domain-invariant perception and expressive action modeling reinforce each other.

Our evaluation is limited to a single platform in a chemistry laboratory setting. Remaining failures trace to limited viewpoint diversity in training renders (perception) and implicit bimanual coordination within a shared action chunk (control). Enriching the rendering distribution and introducing explicit contact-conditioned coordination are concrete next steps; extending GEARS to diverse embodiments and task domains is a broader promising direction.